

Finding Zeros of Quadratic Functions Graphically

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Zero of a Quadratic Function

A **zero** (or **x-intercept**) of a quadratic function is the value of x where the graph crosses or touches the x -axis.

Zeros as Solutions

The **zeros** are the solutions of the equation obtained by setting $y = 0$.

$$y = 0$$

We solve this equation to find the zeros.

Factored Form

A quadratic function in **factored form**,

$$y = a(x - r_1)(x - r_2)$$

has zeros at

$$x = r_1 \quad \text{and} \quad x = r_2$$

Standard Form

A quadratic function in **standard form**,

$$y = ax^2 + bx + c$$

may require graphing first before identifying its zeros.

Two Real Zeros

If the graph **crosses** the x -axis at **two points**, the function has **two real zeros**.

The graph looks like a "U" that cuts through the x -axis twice.

One Repeated Real Zero

If the graph **touches** the x -axis at exactly **one point** (the vertex), the function has **one repeated real zero**.

The graph looks like a "U" that just kisses the x -axis.

No Real Zeros

If the graph **does not intersect** the x -axis, the function has **no real zeros**.

The entire graph stays above or below the x -axis.

Summary: Zeros at a Glance

Graph Behavior	Number of Zeros
Crosses x -axis twice	Two real zeros
Touches x -axis once	One repeated zero
Never touches x -axis	No real zeros

Examples

Example 1: Two Real Zeros

Find the zeros of

$$y = (x - 1)(x - 4)$$

Setting $y = 0$:

$$x = 1 \quad \text{and} \quad x = 4$$

The graph crosses the x -axis at $x = 1$ and $x = 4$.

Example 2: One Repeated Zero

Find the zero(s) of

$$y = (x + 2)^2$$

Setting $y = 0$:

$$x = -2$$

The graph touches the x -axis only at $x = -2$.

Example 3: No Real Zeros

Find the zero(s) of

$$y = x^2 + 1$$

Since $x^2 + 1$ is always positive, the graph never crosses the x -axis.

This function has **no real zeros**.

Practice Exercises

Practice 1

Graph the function. Then determine its zero(s) from the graph.

$$y = (x - 2)(x - 5)$$

Practice 2

Graph the function. Then determine its zero(s) from the graph.

$$y = (x + 4)(x - 1)$$

Practice 3

Graph the function. Then determine its zero(s) from the graph.

$$y = (x - 3)^2$$

Practice 4

Graph the function. Then determine its zero(s) from the graph.

$$y = x^2 - 5x + 6$$

Practice 5

Graph the function. Then determine its zero(s) from the graph.

$$y = x^2 + x - 6$$

Practice 6

Graph the function. Then determine its zero(s) from the graph.

$$y = x^2 - 4$$

Practice 7

Graph the function. Then determine its zero(s) from the graph.

$$y = x^2 - 2x - 3$$

Practice 8

Graph the function. Then determine its zero(s) from the graph.

$$y = x^2 + 2x + 1$$

Practice 9

Graph the function. Then determine its zero(s) from the graph.

$$y = x^2 - 6x + 8$$

Practice 10

Graph the function. Then determine its zero(s) from the graph.

$$y = x^2 + 5x + 4$$

Activities

Activity 1

Graph the function. Then determine its zero(s) from the graph.

$$y = (x - 1)(x - 6)$$

Activity 2

Graph the function. Then determine its zero(s) from the graph.

$$y = (x + 2)(x - 5)$$

Activity 3

Graph the function. Then determine its zero(s) from the graph.

$$y = (x + 3)^2$$

Activity 4

Graph the function. Then determine its zero(s) from the graph.

$$y = x^2 - 7x + 12$$

Activity 5

Graph the function. Then determine its zero(s) from the graph.

$$y = x^2 - x - 6$$

Activity 6

Graph the function. Then determine its zero(s) from the graph.

$$y = x^2 - 9$$

Activity 7

Graph the function. Then determine its zero(s) from the graph.

$$y = x^2 + x - 6$$

Activity 8

Graph the function. Then determine its zero(s) from the graph.

$$y = x^2 - 6x + 9$$

Activity 9

Graph the function. Then determine its zero(s) from the graph.

$$y = x^2 - 8x + 15$$

Activity 10

Graph the function. Then determine its zero(s) from the graph.

$$y = x^2 + 7x + 10$$

Thank you!